

TX PULSE

TX PULSE

EU868 Ref: TX PULSE 600-036

US915 Ref: TX PULSE 600-136

This sensor provides wireless remote meter reading for your pulse meters (dry contact or open collector). It counts pulses and periodically transmits the index by radio. Compatible with LoRaWAN and LoRa Easy modes (Enless proprietary mode), its data is sent by radio without cable pulling or heavy work and can be processed on your BMS or your monitoring tools.



General information

About the TX PULSE



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Activating the TX PULSE



Using the sensor

Using the TX PULSE



Troubleshooting

Frequently asked questions

General information

TX PULSE: general information





Designed for remote meter reading, this LoRaWAN sensor connects to your pulse meters (dry contact or open collector). It counts the number of pulses generated by the meter and periodically sends the index by radio.

Compatible with the LoRaWAN and LoRa Easy protocols (Enless proprietary protocol), its data can be sent directly to your supervision system or your BMS, without cable runs or major work.

Radio

Supported protocols	LoRaWAN LoRa Easy (Enless proprietary protocol)
LoRaWAN class	Class A
LoRaWAN version	Version 1.0.2 RP001-1.02. revB
Frequency	EU868
Identifiers	LoRaWAN: DevEUI, AppEUI, AppKey LoRa Easy: LoRa ID
Transmission power	25 mW (14 dBm)
Encryption	AES128 in LoRaWAN mode

Features

Enclosure protection rating	IP65
Mounting	Wall-mounted (screwed), clamp, DIN rail
Operating temperature	-20 / +55°C
Power supply	3.6V Lithium D-type battery (BAT D 1000-031) Info: Replaceable battery
Battery life	12 years for one transmission every 5 minutes (SF9) Up to 15 years of autonomy Warning: prolonged storage of batteries without use (more than 6 months) may lead to a risk of passivation and a reduction in the advertised autonomy.
Configuration	LoRaWAN: Downlink LoRa Easy: From the receiver server

Measurements

Interface	Pulse inputs
Number of inputs	2 inputs
Input type	Dry contact and open collector
Supported pulse types	Duration: 50 ms minimum Frequency: 10 Hz maximum
Additional function	Debounce

Dimensions

 The warranty for this product is 24 months

Activation procedure

TX PULSE: activation procedure

The TX PULSE can operate in LoRaWAN mode but also in LoRa Easy mode (Enless proprietary mode). In LoRaWAN mode, it can communicate on a public or private LoRaWAN network (including with the Enless LoRaWAN gateway). In LoRa Easy mode, it can communicate with an Enless LoRa Easy receiver.

Thus, there are three possible types of installations:



LoRaWAN Gateway

Installation with the LoRaWAN gateway

[RX LORAWAN](#)



LoRa Easy Receivers

Installation with a LoRa Easy receiver

[RX MODBUS](#)

[RX BACNET](#)

[RX HTTP](#)



Other LoRaWAN network

Installation on a LoRaWAN network

[PUBLIC OR PRIVATE](#)

Enless LoRaWAN gateway

TX PULSE: Enless LoRaWAN gateway

Let's see here how to use your TX PULSE with the **Enless LoRaWAN Gateway**.

 If you have not yet gotten to grips with your **LoRaWAN Gateway**, please first consult the [installation guide](#) for the LoRaWAN gateway (ref: RX LORAWAN 500-602)

Register the sensor

Log in to the configuration interface of your **LoRaWAN Gateway** and go to the **TX CONFIG**.

When manually adding a sensor, the following information must be entered:

1. **Profile** : Select the corresponding Enless sensor model (`600-036 | TX PULSE`).
The profile choice automatically adjusts the decoding options.
2. **DevEUI** : 16-character hexadecimal unique identifier specific to the sensor (available on the product label and in the QR code).
3. **AppKey** : 32-character hexadecimal security key required for LoRaWAN pairing (available on the product label and in the QR code).
4. **Location** : Free text field allowing you to physically identify the sensor's location in the building (e.g. `Technical room - ground floor`).
5. **Interval** : Defines the desired periodicity in minutes between two data transmissions (*uplinks*).
6. **Advanced Settings** : Depending on the chosen profile, this section unlocks specific parameters (alert thresholds, special operating mode, etc.).

Finish adding the sensor by clicking the **Add**.

 The **activation keys of your LoRaWAN sensors** Enless can be provided by the **format .csv** by our **Administration from the Sales**.
Please **contact** adv@enless.fr

Once the sensor has been added, it should appear in the list of Enless sensors.

Check that your sensor is indeed in LoRaWAN mode



LoRaWAN mode is enabled by a jumper located on the main board. To activate this mode, the jumper must be in **position open** (not connecting the two pins), as shown opposite.

- i** The **switching** of the switch jumper is not **taken into account** until the product is **powered off**. Make sure your product is completely powered off before setting the jumper.

Getting started

Only a few steps remain to get the TX PULSE up and running :

1

Position the sensor at its final location

It is important that you power the sensor in the place where it will be permanently installed.

Why is that? Once powered, the sensor will settle on a [Spreading Factor ↗](#) appropriate to its distance from the gateway. If you power your sensor too close to the gateway, then move it away to install it in its final location, it may keep a low Spreading Factor, and as a result communication may no longer be established once it is moved away from the gateway. Ideally, the sensor should lock onto the correct Spreading Factor as soon as it is installed.

2

Power supply

Power the TX PULSE by plugging in the battery Molex connector.

3

Activation

The TX PULSE will proceed with its activation with the *gateway* Enless; the status LED will flash **green** then flash one of the following three colors for 30 seconds:

- **Green** : The sensor is activated.
- **Orange** : The sensor is activated but the RSS is low, so there is a risk of losing LoRa frames.
- **Red** : The sensor is not activated, it needs to be moved closer to the *gateway*.

4

Operation

After activation, the LED flashes let you monitor the sensor status:

Green	Flash every minute
Orange	Flash with each frame transmission Stays solid in case of low battery
Red	Flash every 1 min in case of alarm

5

Installation

Once the TX PULSE has been properly activated on the network, all that remains is to mount it on the wall:

- By fixing it to the wall (screws and wall plugs not provided).
- By mounting it on the DIN rail (mounting accessory not provided)

Network connectivity management

TX sensors include advanced connectivity management mechanisms to ensure reliable and optimized communications on the LoRaWAN network.

Join Requests (Activation procedure)

- **Initialization** : Sensors automatically initiate a *Join* as soon as they are powered on.
- **Link check** : In normal operation, the sensor sends a check request (LinkCheckReq) every 24 hours.
- **Link maintenance** : To confirm that the session is valid, the sensor must receive LinkCheckAck responses from the network.
- **Automatic reactivation** : In case of prolonged failure (if 6 *LinkCheckReq* consecutive requests go unanswered), the sensor considers the session lost and initiates a new *Join*.

ADR (Adaptive Data Rate)

- **Initial configuration** : During the first *Join*, the transmitter uses SF12 and BW125 parameters by default.
- **Dynamic optimization** : The sensor keeps these maximum-robustness parameters as long as it does not receive a LinkADRReq command from the network server (via the gateway) instructing it to adapt its *Spreading Factor* to optimize power consumption and bandwidth.

Go to the section [Usage](#) to learn more about the sensor's installation and features.

Enless LoRa Easy receivers

TX PULSE: LoRa Easy Enless receivers

Let's see here how to use your TX PULSE with a receiver **LoRa Easy Enless** :

- RX MODBUS 500-302
- RX BACNET 500-322
- RX HTTP 500-322

! If you have not yet gotten to grips with your **LoRa Easy receiver**, please first consult the corresponding installation guide.

[RX MODBUS](#)

[RX BACNET](#)

[RX HTTP](#)

Register the sensor

Log in to the configuration interface of your **LoRa Easy receiver** and go to the **TX CONFIG**.

When manually adding a sensor, the following information must be entered:

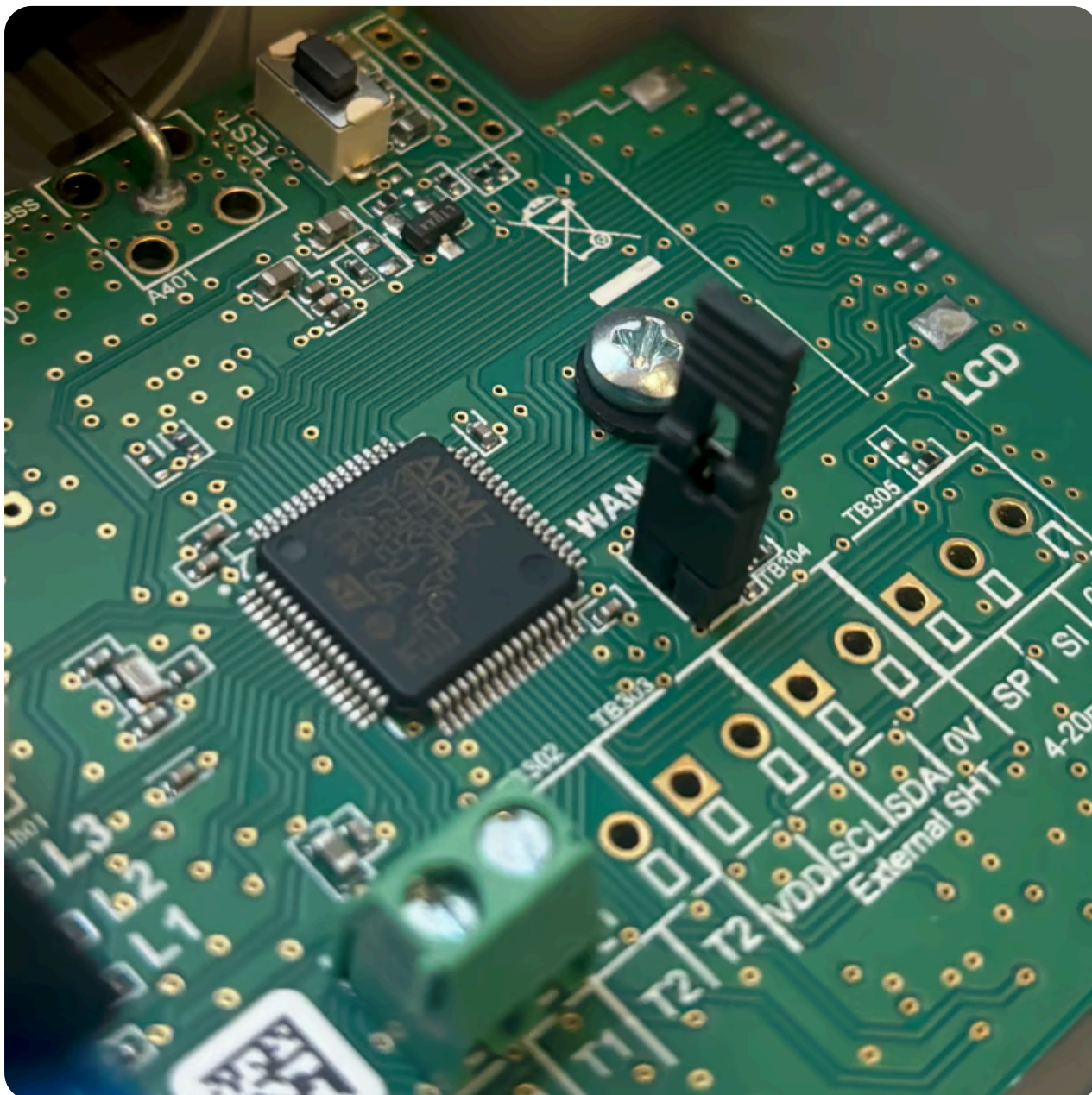
1. **TX Type** : Enless sensor model selection (**600-036 | TX PULSE**) compatible with LoRa Easy.
2. **Location** : Free field to identify the physical location of the sensor (e.g., **Technical room - ground floor**).
3. **LoRa ID** : Unique device identifier, available on the label attached to the sensor housing.
4. **Periodicity** : Time interval (in minutes) between each data transmission to the gateway.

In addition to the previous parameters, optional parameters are available via the button **+ Options** :

1. **Update config** : If enabled, the gateway will automatically synchronize the configuration with the sensor every 24 hours.
2. **Specific options** : Depending on the chosen TX type, additional parameters related to the sensor's specific features may appear.

Finish adding the sensor by clicking the **Add**.

Check that your sensor is indeed in LoRa Easy mode



LoRaWAN mode is enabled by a jumper located on the main board. To activate this mode, the jumper must be in **position closed** (connecting the two pins), as shown opposite.

i The **switching** of the switch jumper is not **taken into account** until the product is **powered off**. Make sure your product is completely powered off before setting the jumper.

Getting started

Only a few steps remain to get the TX PULSE up and running :

1

Power supply

Power the TX PULSE by connecting the battery Molex connector.

2

Activation

The TX PULSE will proceed with its activation with the *of the LoRa Easy receiver* Enless; the status LED will flash **green** then for 30 seconds take one of the following three colors:

- **Green** : The sensor is activated.
- **Orange** : The sensor is activated but the RSSI is weak, so there is a risk of LoRa frame loss.
- **Red** : The sensor is not activated. Several possible reasons :
 1. Check that the jumper is in the LoRa Easy position (see above).
 2. Check that you have entered the sensor's LoRa ID on the receiver
 3. The sensor is too far from the receiver.

In the tab **NETWORK**, the RSSI value allows you to validate the quality of the signal with the receiver:

Good signal

Up to -105 dBm



Adequate signal

from -106 to -112 dBm



Weak signal

above -112 dBm



 In the event of poor RSSI, we **recommend** installing a **repeater of signal**.
To learn more about installing the signal repeater, please visit [the RX REPEATER installation guide](#).

3

Operation

After activation, the LED flashes let you monitor the sensor status:

Green	Flash every minute
Orange	Flash with each frame transmission Stays solid in case of low battery
Red	Flash every 1 min in case of alarm

4

Installation

Once the TX PULSE is properly activated on the network, all that remains is to install it on the wall:

- By fixing it to the wall (screws and wall plugs not provided).
- By mounting it on the DIN rail (mounting accessory not provided)

Go to the section [Usage](#) to learn more about the sensor's installation and features.

Other LoRaWAN networks (public or private)

TX PULSE: other LoRaWAN networks (public or private)

Your TX PULSE is compatible with networks **public LoRaWAN** and **private**. To make full use of it, here are the elements you will need.

Register the sensor

Activation

Your transmitter is activated via Over-The-Air (**OTAA**) only thanks to its three identification keys.

DevEUI	Unique identifier, 64 bits (8 bytes / 16 hexadecimal characters). e.g.: <code>00:18:B2:00:00:00:00:01</code>
AppEUI	Global identifier designating the entity responsible for activation, 64 bits (8 bytes / 16 hexadecimal characters). e.g.: <code>70:B3:D5:7E:D0:00:AF:12</code>
AppKey	Symmetric encryption key, 128 bits (16 bytes / 32 hexadecimal characters). e.g.: <code>2B7E151628AED2A6ABF7158809CF4F3C</code>



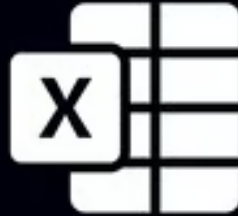
Where can I find the keys?

The LoRaWAN keys for Enless sensors are available:

- By scanning the **QR code** attached to the sensor
- On **the back label** of the sensor
- On **request** from our service department **Administration from the Sales** (adv@enless.fr )

Decoding

Enless Wireless provides various **codecs** to decode the *uplinks* of your transmitter as well as generate *downlinks* to send to it.



Excel file for decoding frames



JavaScript Codecs



TTN Codecs



Chirpstack Codecs



Multitech AP Conduit BACnet Codecs

Check that your sensor is indeed in LoRaWAN mode



LoRaWAN mode is enabled by a jumper located on the main board. To activate this mode, the jumper must be in **position open** (not connecting the two pins), as shown opposite.

i The **switching** of the switch jumper is not **taken into account** until the product is **powered off**. Make sure your product is completely powered off before setting the jumper.

Getting started

Only a few steps remain to get the TX PULSE up and running :

1

Power supply

Power the TX PULSE by plugging in the battery Molex connector.

2

Activation

The TX PULSE will proceed with its activation on the LoRaWAN network; the status LED will flash and then for 30 seconds take on one of the following three colors:

- **Green** : The sensor is activated.
- **Orange** : The sensor is activated but the RSSI is weak, so there is a risk of LoRa frame loss.
- **Red** : The sensor is not activated. Several possible reasons :
 1. The switch was not in the LoRaWAN position at the time of activation
 2. The DevEUI, AppEUI and AppKey keys were not entered correctly
 3. The LoRaWAN network does not adequately cover the area

3

Operation

After activation, the LED flashes let you monitor the sensor status:

Green	Flash every minute
Orange	Flash with each frame transmission Stays solid in case of low battery
Red	Flash every 1 min in case of alarm

4

Installation

Once the TX PULSE has been properly activated on the network, all that remains is to mount it on the wall:

- By fixing it to the wall (screws and wall plugs not provided).
- By mounting it on the DIN rail (mounting accessory not provided)

Network connectivity management

TX sensors include advanced connectivity management mechanisms to ensure reliable and optimized communications on the LoRaWAN network.

Join Requests (Activation procedure)

- **Initialization** : Sensors automatically initiate a *Join* as soon as they are powered on.
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- **Automatic reactivation** : In case of prolonged failure (if 6 *LinkCheckReq* consecutive requests go unanswered), the sensor considers the session lost and initiates a new *Join*.

ADR (Adaptive Data Rate)

- **Initial configuration** : During the first *Join*, the transmitter uses SF12 and BW125 parameters by default.
- **Dynamic optimization** : The sensor keeps these maximum-robustness parameters as long as it does not receive a *LinkADRReq* command from the network server (via the gateway) instructing it to adapt its *Spreading Factor* to optimize power consumption and bandwidth.

Go to the section [Usage](#) to learn more about the sensor's installation and features.

Use

TX PULSE: operation



Use of the button push button

The push button provides access to multiple functions.



Power supply

Operates exclusively with the supplied battery.



Installation of the sensor

Install your TX correctly in its environment.



Wiring

Wire your TX PULSE according to your use cases.



Pulse counting

More information on pulse counting

Push-button use

TX PULSE: push-button operation

The TX PULSE sensor is equipped with a push-button intended mainly for installation and diagnostic phases.

Sending test frames

The push button has only one function:

- **Action** : Briefly press (single push) the button.
- **Result** : The sensor immediately forces the sending of a test frame on the LoRa network.



This manual transmission makes it possible to instantly verify the signal range and the reception of data in your application, without waiting for the next automatic measurement interval.

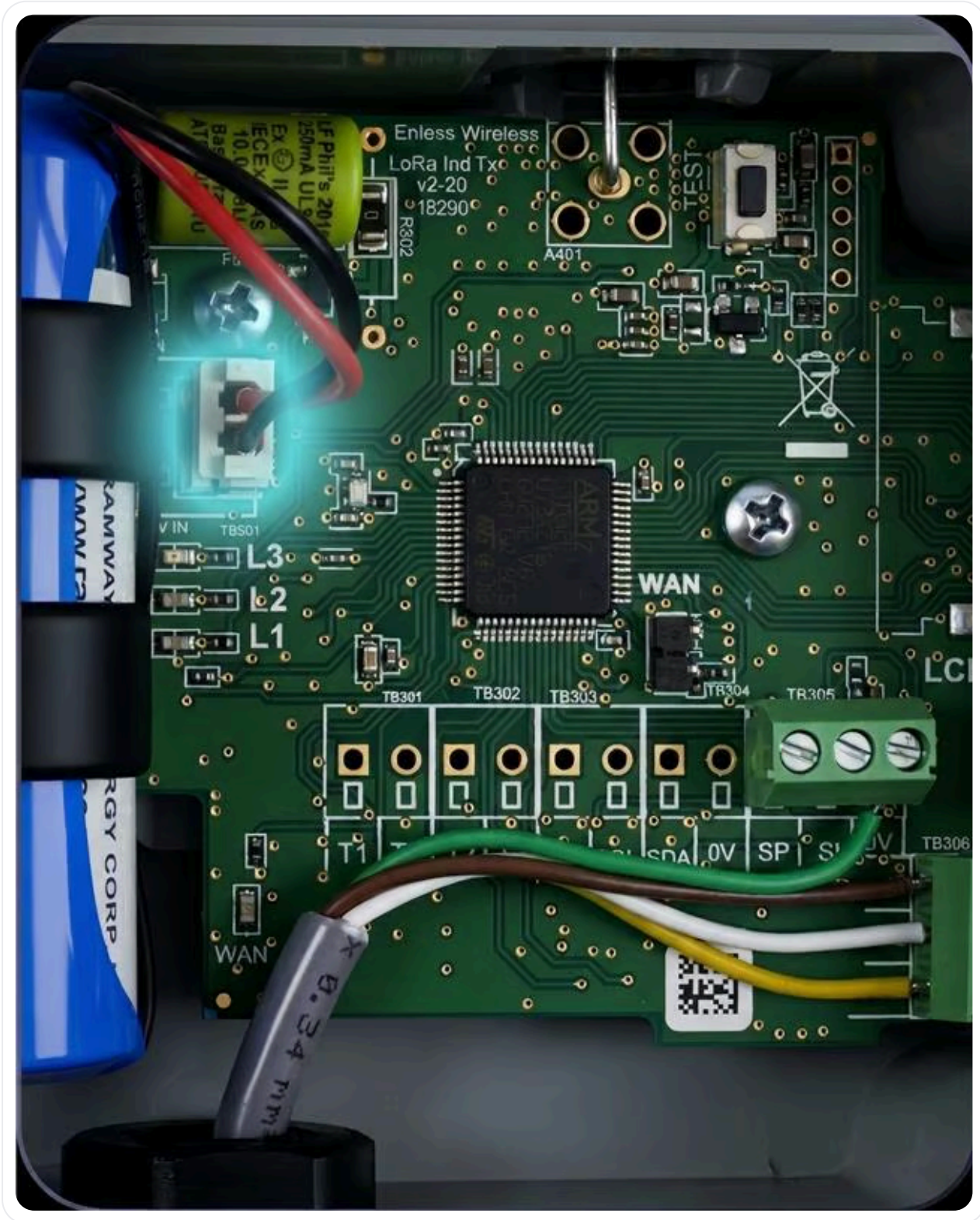
Power supply

TX PULSE: power supply

The TX PULSE offers a power supply mode on **Battery**; use the battery provided for optimal battery life.

First startup

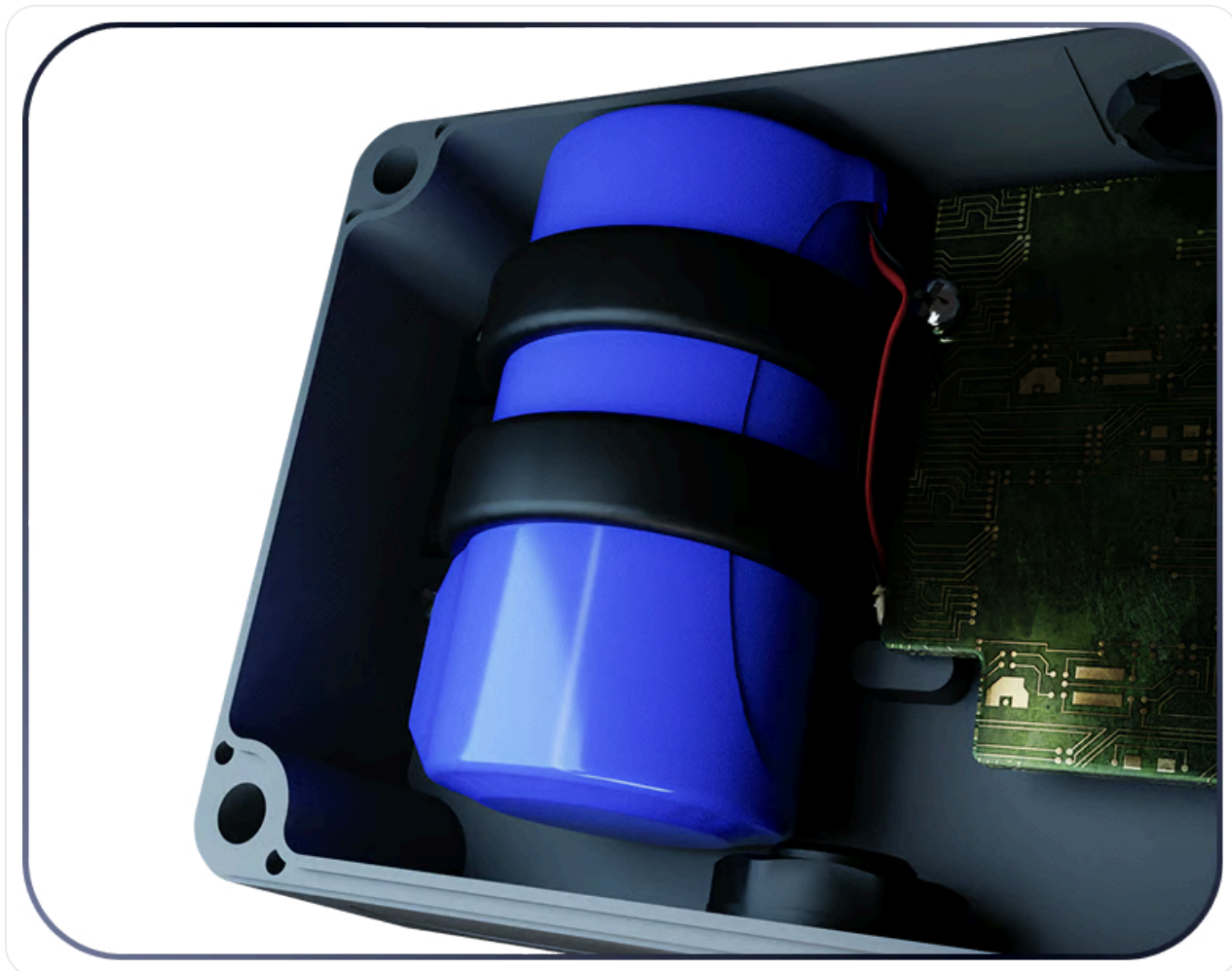
Battery



1. First, unscrew the 4 screws on the front of the sensor.

2. The TX PULSE power is activated by plugging in the Molex connector.

Battery replacement



1. Disconnect the Molex connector of the used battery from the PCB connector.
2. Unscrew the screws on the battery retaining clip and remove it.
3. Remove the used battery from its slot and insert the new one.
4. Put the battery retaining clip back in place and lock it using the two screws provided for this purpose.
5. Reconnect the Molex connector of the new battery to the PCB.
6. The sensor restarts.



Be careful **not to touch** the **components** of the PCB with your fingers when handling the battery connector.

-  Do not close the sensor right away so that you can check the LED sequence indicating its activation on the LoRa Easy or LoRaWAN network.

Sensor installation

TX PULSE: sensor installation



The TX PULSE offers three mounting solutions to suit all installation configurations:

- Wall mounting
- DIN rail mounting
- Cable tie mounting

Wall Mounting

The enclosure has two external mounting points, eliminating the need to open the cover for installation:

- Locations: A mounting hole is located on the top and a second one on the bottom of the enclosure.

- **Method:** Use screws and anchors suitable for the type of support you have (solid wall, drywall, etc.).

i The screws are not supplied; make sure to use screws and anchors suitable for the type of support you have (plasterboard, concrete, wood) to ensure the sensor is held securely.

DIN Rail Mounting (Optional)

For a neat integration into an electrical cabinet on a standard 35 mm DIN rail, you can use the specific Enless Wireless accessory.

- **Reference:** **DIN RAIL 1000-005.**
- **Packaging:** Sold in pairs (two clips per receiver).
- **Installation:**
 - 1. **Screw** the clips to the back of the TX enclosure.
 - 2. **Clip** the assembly simply onto the 35 mm DIN rail. The mechanism allows for quick and secure locking without additional tools.

Cable Tie Mounting

Flexible solution for mounting on pipes, poles, or temporary structures.

- **Installation :** The frame has two cable tie pass-throughs on each side (left and right). This configuration allows the sensor to be held firmly using standard cable ties (Rilsan type).

Wiring

TX PULSE: wiring

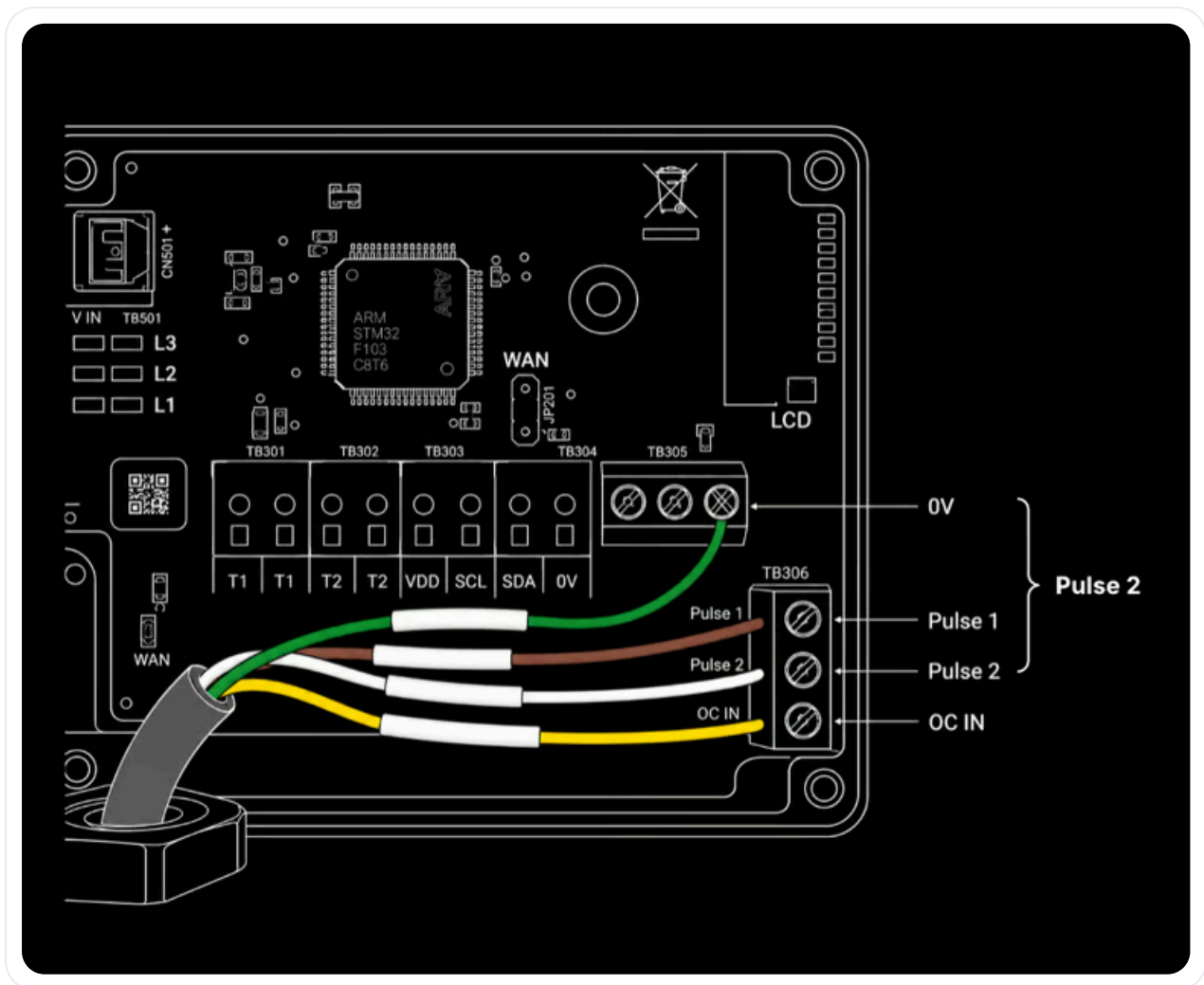
The TX PULSE wiring adapts to your installation according to three possible configurations: you can use the input **Pulse 1** or **Pulse 2** for connecting standard dry-contact meters, or choose the connection in **Open Collector** (open collector). Whichever mode is chosen, the sensor ensures accurate counting of the impulses generated by your meter for reliable transmission of the index readings.

-  The TX PULSE sensor is compatible with meters with pulse outputs in dry contact or open collector.

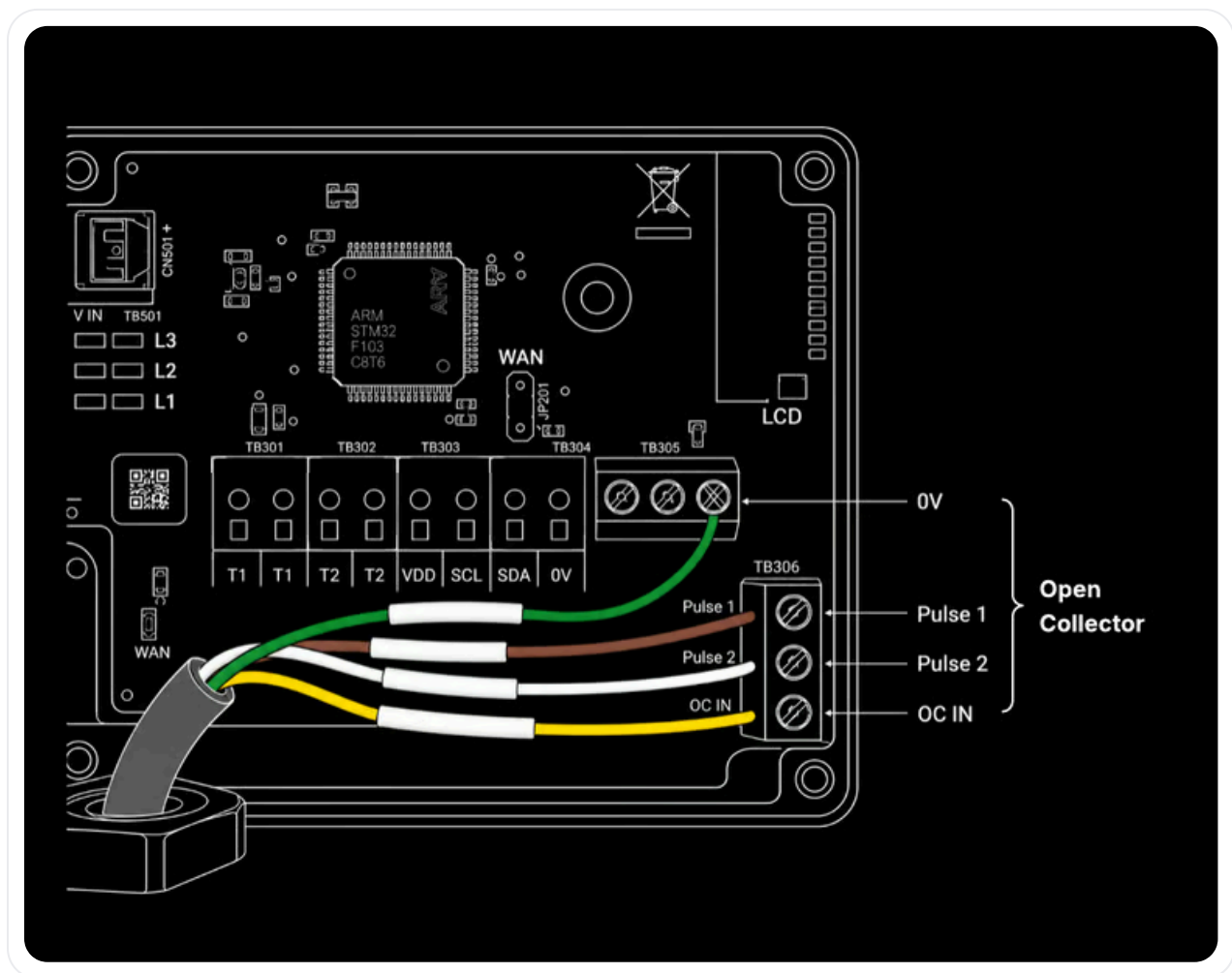
Pulse duration = 50 ms minimum **Pulse frequency** = 10 Hz maximum

The diagram illustrates the internal components and connections of the STM32F103C8T6 development board. Key components include the ARM STM32F103C8T6 microcontroller, a USB connector (CN501), a V IN TB501 header, and various test points (TB301-TB305). The Pulse 1 module is connected to the board via a ribbon cable. The connections are labeled: Pulse 1 (green wire) to TB305, Pulse 2 (brown wire) to TB304, and OC IN (yellow wire) to TB306. The board also includes a WAN header, a QR code, and a LCD display.

Pulse 2



Open collector



⚠ For pulse transmitters, we guarantee **proper operation** with a **cable** that can extend **up to 3 meters**.

Pulse counting

More information on pulse counting

The TX PULSE pulse sensor is compatible with dry-contact and open-collector pulse meters. Make sure your meter is indeed outputting pulses that meet the following parameters:

- 50 ms minimum pulse duration
- 10 Hz maximum pulse frequency



The TX PULSE sensor counts and periodically transmits an increasing cumulative number of pulses, so that any possible loss of radio communication does not disrupt your counting. Pulses counted during one period are added to those counted during the previous period. The counting is independent of the radio part; if a radio frame is not received, you will receive the correct index in the next frame.

Troubleshooting

TX PULSE: troubleshooting

Troubleshooting

Need help with your TX PULSE ATEX? Access our FAQ below or ask your question to the assistant



Ask your question

Go to FAQ

↗ Frequently Asked Questions

✓ **Maximum number of pulses**

The sensor can record a maximum of 4,294,967,295 pulses

✓ **Maximum pulse cable length**

We recommend not exceeding 3 meters of cable.

✓ **Reset the index to zero**

To reset the pulse count for each input, please power the transmitter off and then back on.

✓ **I can't connect my sensor to my LoRa Easy receiver**

If you are unable to connect your sensor to your LoRa Easy receiver, first check that the jumper or switch on the product is properly set to LoRa mode before powering the sensor.

If not:

1. Power off your sensor
2. Set the switch or jumper to LoRa mode
3. Power the sensor back on



Also make sure that the sensor's LoRa ID has been correctly declared on your receiver.

✓ **I have a fixed orange LED on my sensor, what does that mean?**

The fixed orange L2 LED is characteristic of a low battery level. We invite you to try powering your transmitter with a newer battery in order to confirm this hypothesis.

✓ How do I decode the battery status of my sensor in the LoRa Easy receiver's Modbus table?

The battery status must be decoded in the "Status" register.

Register	Hexadecimal	Decimal	Details
4	0000	0	Device ID HI
5	0110	272	Device ID LO
6	0201	513	Tx Type 2 / Version 1
7	0004	4	Status
8	0000	0	Alarm Status
9	00c5	197	Temperature 19.7 °C
10	0187	391	Humidity 39.1%
11	0004	4	VOC 4 ppb
12	01f1	497	CO2 497 ppm
13	0000	0	
14	0000	0	
15	0000	0	
16	0000	0	
17	0000	0	
18	0000	0	
19	0000	0	
20	0000	0	
21	0032	50	Transmission Counter 50
22	fff6	-10	Received RSSI value -10 dBm
23	0000	0	0 minutes since last reading

Here, the value read is 0004 (hexadecimal). Convert this hexadecimal value to binary on 16 bits.

0004(hex) = 00000000 0000**0100** (binary)

Bits 3 and 2 **in red** are the ones that provide information about the battery level:

00 = 100%

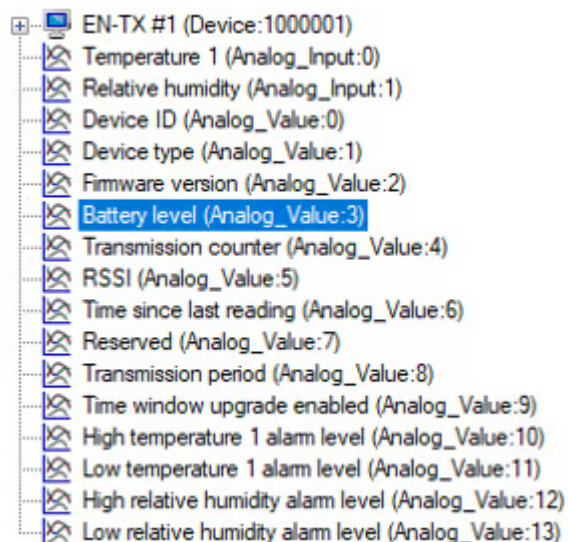
01 = 75%

10 = 50%

11 = 25%

✓ How do I decode the battery status of my sensor in the LoRa Easy receiver's BACnet objects?

The battery status must be read in the "Battery level" object.



The alert level for the battery is given in 4 levels:

0 = 100% remaining

1 = 75% remaining

2 = 50% remaining

3 = 25% remaining (battery replacement to be planned)

✓ I can't connect my sensor to my LoRaWAN network

If you are unable to connect your transmitter to the network, first check that the jumper is properly set to LoRaWAN mode before powering the transmitter. The green WAN LED should flash when you power the transmitter. The jumper positions are:

- **LoRaWAN** : a single pin
- **LoRa Easy** : two pins

! It is essential to declare the transmitter by entering its LoRaWAN keys on your network backend or on your gateway interface **BEFORE connecting the battery**.

✓ **Decoding my sensor frames when I use it in LoRaWAN mode**

To decode frames from an Enless LoRaWAN sensor, please use our decoding assistance tools: <https://enless.gitbook.io/centre-aide/ressources/capteurs/codecs-et-decodage> ↗

✓ **Which port should downlinks be sent on in LoRaWAN?**

Downlinks should be sent on port number 1.

✓ **What is a Spreading Factor?**

The Spreading Factor (SF) in LoRaWAN is used to define how a sensor sends its data. The lower the SF (e.g. SF7), the faster the message is sent, with lower battery consumption, but over a shorter distance. Conversely, a higher SF (e.g. SF12) makes it possible to send data much farther, even with obstacles, but the transmission is slower and consumes more energy. It can be compared to a person speaking quickly to someone nearby, or slowly and clearly to be heard from far away. The choice of SF is therefore a compromise between radio range, battery life, and transmission speed.

✓ How do I change the transmission period of my sensor in LoRaWAN mode?

Changing the transmission frequency of a LoRaWAN transmitter is done by Downlink.

The Excel file below allows you to decode frames and generate configuration Downlinks: <https://enless.gitbook.io/centre-aide/ressources/capteurs/codecs-et-decodage/excel-de-decodage-des-trames> ➔

Example: We want to change the periodicity of a 600-023 transmitter to 15 minutes.

Step 1: Please open the attached Excel file and go to the sheet for the transmitter in question.

Enter LoRaWAN String Below for Analysis:											
000081060E1100D3016500AD026100800000											
Start	Count	HEX VALUE	DEC / BIN								
1	6	000081	129	Transmitter ID							
7	2	06	6	6 = Ambient T&H / VOC / CO2 Sensor							
9	2	0E	14	Sequential Counter							
11	2	11	00010001	F/W (bits 3-0), Settings (bits 7-4)							
13	4	00D3	21,1	Temperature							
17	4	0165	35,7	Humidity							
21	4	00AD	173	VOC							
25	4	0261	609	CO2							
29	4	0080	128	Alarm Status				00000000	10000000	0	0
33	4	0000	0	Status				00000000	00000000	0	0

Below is used to construct the message to re-configure transm											
Generic Field Names		Transmitter Specific Fields (only change those in Green)		Hex		Copy Str					
Fixed		Fixed (0)		0	000000	0000000030000012C000000000					
Installation Packet = 3		Fixed (3)		3	03						
Fixed		Fixed (0)		0	0000						
RE-Tx Time (secs)		Periodicity (mins)		5	012C						
Re-Tx retries		Enter 1 or 0		0	0000						
TWU Period (secs)		TWU (hrs)		0	0000						
P1		Hi Temp Alarm (degC)		30,0	012C						
...		...		...	...						

600-021	600-022	600-023	600-031	600-036	600-037	600-038	600-039	600-232	600-035	600-034	600-032	+
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Step 2: Please change the periodicity and press "Enter" to validate.

Below is used to construct the message to re-configure transmitter via the LoRaWAN network

Generic Field Names	Transmitter Specific Fields (only change those in Green)	Hex	Copy String Below and use to send on LoRaWAN network
Fixed	Fixed (0)	0	000000
Installation Packet = 3	Fixed (3)	3	03
Fixed	Fixed (0)	0	0000
RE-Tx Time (secs)	Periodicity (mins)	15	0384
Re-Tx retries	Enter 1 or 0	0	0000
TWU Period (secs)	TWU (hrs)	0	0000
P1	Hi Temp Alarm (degC)	30.0	012C
P2	Lo Temp Alarm (degC)	15.0	0096
P3	Hi Hum Alarm (%)	50.0	01F4
P4	Lo Hum Alarm (%)	20.0	00C8
P5	VOC Hi Alarm (ppb)	200	00C8
P6	VOC Lo Alarm (ppb)	50	0032
P7	CO2 Hi Alarm (ppm)	650	028A
P8	CO2 Lo Alarm (ppm)	250	00FA
P9	Fixed (0)	0	0000
P10	Fixed (0)	0	0000
P11	Fixed (0)	0	0000
P12	Fixed (0)	0	0000
Fixed	Fixed (1)	1	01

Step 3: A new Downlink taking the new periodicity into account will be generated. Please copy this downlink and paste it into the appropriate field on your gateway. Downlinks should be sent on the **port1**.

Below is used to construct the message to re-configure transmitter via the LoRaWAN network

Generic Field Names	Transmitter Specific Fields (only change those in Green)	Hex	Copy String Below and use to send on LoRaWAN network
Fixed	Fixed (0)	0	000000
Installation Packet = 3	Fixed (3)	3	03
Fixed	Fixed (0)	0	0000
RE-Tx Time (secs)	Periodicity (mins)	15	0384
Re-Tx retries	Enter 1 or 0	0	0000
TWU Period (secs)	TWU (hrs)	0	0000
P1	Hi Temp Alarm (degC)	30.0	012C
P2	Lo Temp Alarm (degC)	15.0	0096
P3	Hi Hum Alarm (%)	50.0	01F4
P4	Lo Hum Alarm (%)	20.0	00C8
P5	VOC Hi Alarm (ppb)	200	00C8
P6	VOC Lo Alarm (ppb)	50	0032
P7	CO2 Hi Alarm (ppm)	650	028A
P8	CO2 Lo Alarm (ppm)	250	00FA
P9	Fixed (0)	0	0000
P10	Fixed (0)	0	0000
P11	Fixed (0)	0	0000
P12	Fixed (0)	0	0000
Fixed	Fixed (1)	1	01

✓ How do I switch from one LoRaWAN network to another?

To switch a sensor from one LoRaWAN network to another, you do not need to physically intervene on the sensor. You only need to remove it from the old LoRaWAN network, then declare it on the new LoRaWAN network.

Indeed, every 24 hours, transmitters send a LinkCheckReq message which must be confirmed by a LinkCheckAck from the network. As long as LinkCheckReq messages are confirmed by a LinkCheckAck, the transmitter will not send a new JOIN. If the transmitters are still declared on the old LoRaWAN network, they remain connected to it. On the other hand, if they are declared on the new LoRaWAN network and are no longer provisioned on the old one, the switch to the new LoRaWAN network takes place.

***JOIN:** this is a LoRaWAN network access phase that allows network parameters to be dynamically renewed between the end device and the LoRa Server.

